Supplementary Material
SplitGuard-AD: Executable Leakage Auditing for Slice-Based Alzheimer's Disease MRI Deep Learning Benchmarks

S1 Consolidated training hyperparameters

Table S1 lists the full per-cohort training configuration referenced in the main text.

Table S1: Consolidated training hyperparameters across the three tiers and both backbones.

Parameter	Kaggle	OASIS-1	ADNI1
Backbones	ResNet-18; DenseNet-121	ResNet-18; DenseNet-121	ResNet-18 primary; DenseNet-121 architecture arm
Initialisation	ImageNet	ImageNet	ImageNet
Input	$3\times224\times224$	$3\times224\times224$	$3\times224\times224$
Normalisation	ImageNet mean/std	ImageNet mean/std	ImageNet mean/std
Augmentation	flip + $\pm7^\circ$	flip + $\pm7^\circ$	flip + $\pm7^\circ$
Epochs	30	20	15 (all arms)
Optimiser	AdamW	AdamW	AdamW
Learning rate	3×10^{-4}	3×10^{-4}	10^{-4}
LR schedule	Cosine anneal	Cosine anneal	Cosine anneal
Weight decay	10^{-2}	10^{-2}	10^{-2}
Batch size	32	32	16
Loss	inverse-freq BCE	inverse-freq BCE	inverse-freq BCE
Seed array	$\{0, 1, 2, 3, 42\}$	$\{0, 1, 2, 3, 42\}$	$\{0, 1, 2, 3, 4\}$
Backend	CUDA H100	CUDA H100	CUDA H100; RTX PRO 4500 (dose-response matrix); RTX 2000 Ada (volumetric arm); RTX 4090 (exact-linkage, structured-corruption and label-permutation arms)
Mixed precision	AMP	off	off

Every arm ran in one code state on one GPU. Mixed precision is the one setting that differs by cohort: the Tier-1 trainer enables automatic mixed precision on CUDA and the OASIS-1 and ADNI trainers do not. Within any one cohort every protocol is trained under identical settings, which is the condition the inflation-gap comparison requires. The volumetric ADNI arm uses a 3D ResNet-18 on 128^3 skull-stripped, z-scored volumes, batch size 4, AdamW at 10^{-4} with weight decay 10^{-2}, 15 epochs. The exact-linkage, structured-corruption and label-permutation arms use the ADNI1 column unchanged; they vary the manifest or the labels, never the training configuration, which is what lets their contrasts be read against the primary arm.

S2 Operating-point sensitivity table (ADNI)

The clinical translation in the main text is anchored at a screening specificity of 0.90.

The cost-of-leakage arithmetic in full The main text quotes one figure from this translation. The identity is deterministic, missed/1000 = $1000 \cdot \text{prev} \cdot (1 - \text{sens})$, applied to the measured sensitivity at a fixed specificity of 0.90 on the converter-inclusive arm. Table S2 reports it at two prevalence anchors rather than one so that the dependence on prevalence is visible rather than hidden in a single number.

Table S2: Cost-of-leakage arithmetic at two prevalence anchors, converter-inclusive ADNI arm, sensitivity read at fixed specificity 0.90. "Leaky apparent" is what the benchmark would predict; "honest actual" is what the component-safe sensitivity delivers at the same prevalence. The difference is the cost of trusting the benchmark. These are prevalence-weighted re-expressions of a measured sensitivity difference, not deployment estimates.

Prevalence anchor	**Leaky apparent**	**Honest actual**	**Additional missed**
	missed AD diagnoses per 1,000 screened		
US population, 75–84, 13.8% [1]	18.5	64.2	45.7
Memory clinic, 58.9% [2]	79.0	274.0	195.0

The two rows differ by a factor of four, which is the point: the count is a function of the population it is applied to, and quoting one anchor without the other would invite exactly the deployment reading the main text declines.

Table S3 sweeps that choice, reporting sensitivity at four fixed specificities, specificity at three fixed sensitivities, and the Youden-J point, for each of the three protocols and for the leaky-minus-honest difference. The anchor is not load-bearing: the leaky-minus-honest difference is positive at every one of the ten operating points in the sweep, from +0.099 (specificity at the Youden point) to +0.424 (sensitivity at specificity 0.95), so no conclusion in the main text depends on where the threshold is placed. The magnitude does vary by roughly fourfold across the sweep, which is why the main text quotes one anchor explicitly rather than presenting the difference as a single number.

S3 Mapping SplitGuard-AD outputs to standard reporting domains

Table S4 maps SplitGuard-AD's pre-training audit, model audit, and reporting artefacts to the standard reporting domains covered by TRIPOD+AI and PROBAST domain 4 (Analysis). We deliberately list domains rather than guideline item numbers because TRIPOD+AI numbering is sensitive to the model-development vs. model-validation arm of the guideline; the substantive mapping stays valid regardless of which arm is being checked. SplitGuard-AD is not a replacement for the broader guidelines; it operationalises the subset concerning data partitioning, reproducibility, calibration, and discrimination as machine-checkable outputs.

Domains not directly operationalised by SplitGuard-AD (title/abstract, background and objectives, outcome/predictor variable definition, model specification) remain author- and reviewer-side checklist work; SplitGuard-AD complements rather than substitutes for the broader guideline. PROBAST domain 4 (Analysis) risk-of-bias items concerning sample handling, missing data, and overoptimism in performance estimation are addressed by the same pre-training audit outputs above.

Table S3: Operating-point sensitivity table for the ADNI converter-inclusive arm. **Every threshold is selected on the validation partition and applied to test**; no operating point is chosen by scanning the test ROC. The fixed-rate rows are mean ± SD across five seeds and the Youden rows are seed means, with Δ the Protocol A minus Protocol C difference.

Operating point	Random	Subject-only	Component-safe	Δ
Sensitivity at fixed specificity				
Spec = 0.80	0.919 ± 0.028	0.717 ± 0.122	0.649 ± 0.084	+0.270
Spec = 0.85	0.901 ± 0.024	0.640 ± 0.109	0.616 ± 0.100	+0.285
Spec = 0.90	0.866 ± 0.046	0.551 ± 0.105	0.535 ± 0.103	+0.331
Spec = 0.95	0.810 ± 0.057	0.382 ± 0.055	0.386 ± 0.099	+0.424
Specificity at fixed sensitivity				
Sens = 0.85	0.948 ± 0.028	0.567 ± 0.114	0.563 ± 0.108	+0.385
Sens = 0.90	0.898 ± 0.061	0.439 ± 0.124	0.488 ± 0.141	+0.410
Sens = 0.95	0.782 ± 0.125	0.299 ± 0.111	0.403 ± 0.162	+0.379
Youden-optimal threshold (selected on validation)				
J	0.785	0.458	0.533	+0.252
Sens at J	0.836	0.768	0.682	+0.153
Spec at J	0.950	0.690	0.851	+0.099

The same outputs answer the CLAIM [3, 4] items a leakage audit touches, listed here by topic rather than by number because the 2024 revision renumbered them. *Data sources and provenance*: the per-image manifest records where each image came from and which identifiers were available, and for Tier 1 the provenance is recovered by content matching rather than assumed (§2.3.1 of the main text). *Data partitioning*: the split rule is frozen in a versioned manifest, the audit reports participant, session and component overlap for each partition, and the compositional check reports how the partitions differ where they are disjoint. *Reference standard*: the label join is audited, with the share of labels resting on a date-proximity match reported alongside the exact-key share. *Evaluation*: performance is reported per protocol with both image-level and participant-level aggregation, calibration, and sensitivity at a fixed specificity. *Failure analysis and reproducibility*: the identity probe, the provenance-degradation sweep and the released rebuild pipeline cover the failure-mode and reproducibility items.

S4 Dose-response mixed-effects model specification

The dose-response OLS slope in the main text (Results: leakage as a dose) is a descriptive calibration constant. To obtain an inferential quantity that respects the paired structure of the 25 (seed, overlap) observations, we additionally fit the random-intercept model

$$\mathrm{AUROC}_{ij} = \beta_0 + \beta_1 p_{ij} + u_i + \varepsilon_{ij}, \quad u_i \sim N(0, \sigma_u^2),\ \varepsilon_{ij} \sim N(0, \sigma^2),$$

by iterated feasible GLS under compound symmetry (`analyze_dose_response_mixed_effects.py` in the release repository). The fitted variance ratio is $\widehat{\rho} = \widehat{\sigma}_u^2/\widehat{\sigma}^2 = 0.317$ on ResNet-18, confirming non-negligible between-seed variation.

Because the design is *balanced* (every seed contributes exactly one observation at each of the five overlap levels) the GLS and OLS point estimates coincide by construction: $\widehat{\beta}_1 = 0.1251$ under

Table S4: SplitGuard-AD outputs mapped to standard reporting domains.

Reporting domain	**SplitGuard-AD output**	**Main-text section**
Source of data & provenance	Per-image manifest with subject/session/series IDs and preprocessing lineage; leakage-graph edges over subject/source-volume/near-duplicate.	Methods: cohorts
Eligibility & splitting rules	Component-safe split rule with class-stratification target; orphan-component exclusions audited.	Methods: audit instrument
Sample-size handling & inference	Five-seed array with "descriptive of seed-level variability" labelling and hierarchical (subject $\times$ seed) bootstrap as the inferential cross-check.	Results: statistical conventions
Missing-data audit	Pre-training audit flags unparsable filename keys (Tier 1), missing CDR (Tier 2), and unresolved diagnosis (Tier 3).	Methods: cohorts
Statistical methods	Paired-seed bootstrap with direction-preservation count; hierarchical bootstrap as the defensible inferential quantity.	Results: statistical conventions
Calibration	Brier score and expected calibration error reported across protocols; monotone degradation with leakage protection.	Results: ADNI robustness
Discrimination	AUROC (image- and subject-level), sensitivity at fixed specificity, ROC curves with operating-point overlay.	Results: ADNI arm and screening operating point
Subgroup performance	Sex-stratified AUROC with paired-seed bootstrap (exploratory).	Supplementary S6, Figure S1
Clinical-utility translation	Sensitivity at fixed specificity = 0.90; predictive values at two prevalence anchors.	Results: screening operating point
Reproducibility & versioning	Frozen split manifests (seed-byte-reproducible CSV), versioned audit report, GitHub release.	Data and Code Availability
Limitations & bias declaration	Single-cohort headline scope, $n = 5$ seeds, 2D-slice representation, component-marginal CI straddles zero, subgroup small-sample.	Discussion: Limitations
Code & artefact availability	Apache-2.0 release of manifest builders, leakage-graph construction, split generation, audit, probe, and bootstrap scripts.	Data and Code Availability

both. The random-intercept structure therefore does not move the slope here; what it changes is the uncertainty attached to it, and that is where the inferential content lies.

Inference uses a cluster-robust sandwich standard error over seed with the finite-cluster scaling $\frac{G}{G-1} \cdot \frac{N-1}{N-k}$ and a $t(G-1)$ critical value. With $G = 5$ seeds the correct multiplier is $t_{0.975}(4) = 2.776$, not the normal 1.960; using the normal quantile at this cluster count understates the interval by roughly 40%. The resulting estimates are: ResNet-18 $\widehat{\beta}_1 = 0.1251$, cluster-robust SE 0.0139, 95% CI $[0.0866, 0.1636]$, with a non-parametric seed-cluster bootstrap (B=10,000) giving the comparable interval $[0.1042, 0.1515]$; DenseNet-121 $\widehat{\beta}_1 = 0.0906$, SE 0.0177, 95% CI $[0.0416, 0.1396]$. Both intervals exclude zero, and they overlap substantially, so the two architectures' dose-response slopes are not distinguishable at five seeds.

Spearman ρ on the pooled 25-point (p, AUROC) pairs is 0.941 for ResNet-18 and 0.851 for DenseNet-121; per-seed AUROC is monotonically non-decreasing in overlap in 5 of 5 ResNet-18 seeds and 0 of 5 DenseNet-121 seeds, consistent with the smaller ResNet-vs-DenseNet inflation-gap asymmetry reported in the ADNI arm.

S5 Metadata linkage-integrity artefact

The full scan-to-visit linkage accounting for the ADNI headline cohort is released as a per-scan CSV (`adni_linkage_audit.csv`, one row per scan), produced by `scripts/build_linkage_audit.py`. Columns:

- `scan_uid`: image identifier (not participant-derived);
- `ptid_hash`: PBKDF2-HMAC-SHA256 of the PTID under a secret per-release salt (600,000 iterations). The salt is never written to the released artefacts;
- `acq_date`: acquisition date, retained because the join cannot be audited without it;
- `join_tier`: `exact_viscode` (direct `(RID, VISCODE)` match), `date_proximity` (visit key failed; label taken from the nearest clinical visit within the 180-day tolerance), or `unresolved`;
- `label_source`: the ADNI column the diagnosis came from;
- `diagnosis_group`: resolved CN / MCI / AD / unknown;
- `in_cnad_universe`: whether the scan enters the CN-vs-AD task.

Tier counts on the released cohort: `exact_viscode` 1,695 (77.7%), `date_proximity` 485 (22.2%), `unresolved` 2 (0.1%). The companion summary (`reports/audits/adni/adni_linkage_audit_summary.json`) records the same counts plus the KDF parameters.

We do not record a candidate-visit count or a minimum date-difference per scan: the manifest builder retains only the tier it resolved to, so those quantities are not recoverable without re-running the join with extra instrumentation. We state this rather than describing fields the released file does not contain.

S6 Sex-stratified performance on ADNI (exploratory)

Stratifying test-set AUROC by sex within each protocol (Figure S1) surfaces a female-minus-male AUROC delta of $+0.102$ $[+0.068, +0.136]$ under subject-only and $+0.115$ $[+0.046, +0.181]$ under component-safe SplitGuard-AD that collapses to -0.020 $[-0.055, +0.015]$ under the leaky protocol, where the sign holds in only $8,893/10{,}000$ resamples. We report the two deltas descriptively and do not test their difference. The gap does not appear to be explained by test-set class imbalance: the female AD share is lower than the male share under all three protocols. Sex differences in AD atrophy trajectories and cognitive reserve are documented in the wider literature [5, 6, 7], but per-cell sample sizes here are small ($\sim$ 8–13 subjects per sex $\times$ age cell) and no adjustment for APOE, education, site, or scanner is performed; **this finding is exploratory and hypothesis-generating only**, and it should not be interpreted as evidence of a biological sex effect without external-cohort replication. The relevant point is methodological rather than biological: leaky evaluation would prevent detection of any such effect if one existed.

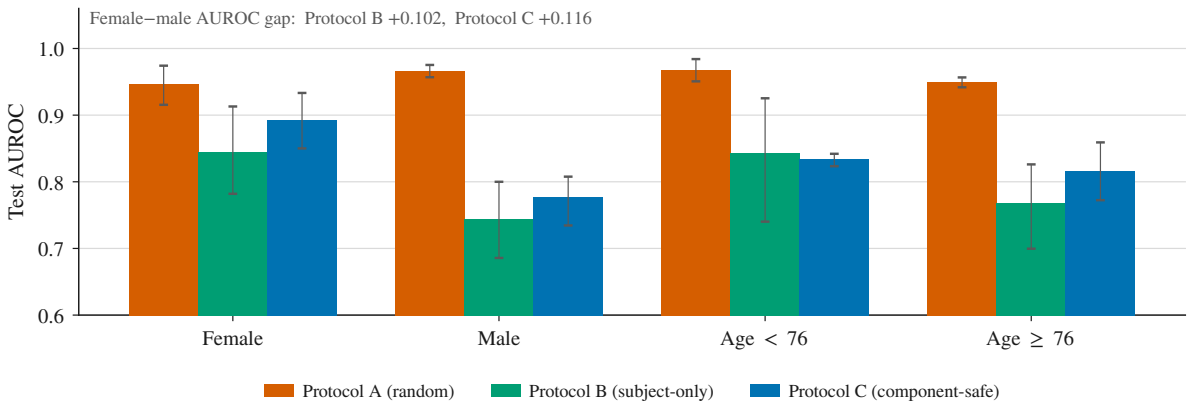


Figure S1: ADNI per-subgroup test AUROC across five seeds (paired-seed bootstrap intervals, $B = 10{,}000$; descriptive). Both honest protocols show a $\sim +0.11$ female-male AUROC gap; under the leaky protocol the same contrast has a point estimate near zero. Reported as exploratory, pending external-cohort replication.

S7 Related work: what each prior treatment of leakage measures

Table S5 places this work against prior treatments of leakage in AD MRI along the six axes the main text uses: whether inflation is shown, whether a dose is measured, whether provenance is varied, whether the mechanism is demonstrated rather than argued, whether damage is observed rather than simulated, and whether split manifests are released.

Table S5: What prior leakage studies establish, and what this paper adds. "Dose" is a measured relationship between the amount of contamination and the reported score, rather than a comparison of two endpoints. "Provenance" is whether the identifiers a protocol needs are treated as a variable rather than assumed. "Mechanism" is whether the inflation is shown to come from patient recall by an intervention that removes the disease signal, rather than argued from saliency or from representation content.

	Inflation shown	Dose measured	Provenance varied	Mechanism shown	Observed damage	Split manifests
Wen *et al.* [8] (review)	✓	–	–	–	–	–
Yagis *et al.* [9]	✓	–	–	–	–	–
Rumala [10]	✓	–	–	saliency	–	–
Rosenblatt *et al.* [11]	✓	three levels	–	–	–	–
Ghiasi *et al.* [12]	✓	–	–	–	–	✓
DataSAIL [13] (tool)	–	–	–	–	–	✓
This paper	✓	continuous	✓	✓	✓	✓

S8 Tier-1 provenance reconstruction: the mapping audit

The main text states that the Tier-1 benchmark is a redistribution of OASIS-1. Because that is a concrete claim about an existing public dataset, it is held to a forensic standard here rather than asserted.

Each Tier-1 image is matched by normalised cross-correlation against every axial slice of every OASIS-1 `t88_masked_gfc` volume. A match is accepted only when one source slice correlates far above every other candidate, so the evidence is the *separation* between the best and second-best match rather than the best match alone. Table S6 reports the audit.

Table S6: Tier-1 to OASIS-1 mapping audit. The decisive row is the separation: the weakest correct match still correlates at 0.9996, while the strongest match to any *other* participant reaches only 0.9623. No threshold between those two values changes any assignment, which is why the mapping is reported as recovered rather than inferred.

Quantity	**Value**
Redistributed images considered	6,400
OASIS-1 volumes searched	436
Images matched	6,400
of which unambiguous	6,400
ambiguous	0
unmatched	0
Source participants recovered	200
Images per recovered participant	32
Axial slice range used	71–102
Participants split across class folders	0
Minimum self-correlation (weakest correct match)	0.9996
Maximum correlation to a different participant	0.9623
Images whose folder label agrees with the source CDR	6,400
Participants whose label agrees with CDR	200 of 200
Recovered participants also present in Tier 2	200 of 200

Two further checks support the reading. The folder labels reproduce the source CDR exactly for all 200 participants, which an accidental or partial correspondence would not do. And every recovered participant is present in Tier 2, so Tier 1 is not an independent cohort and is not counted as one anywhere in the main text.

`scripts/audit_tier1_ground_truth.py` regenerates this table from a reader's own Kaggle and OASIS-1 downloads. The participant-level mapping itself is not redistributed; the script that recovers it is.

S9 Which leakage-graph edge rules fire, per cohort

Table S7 counts, per cohort, how often each edge rule fires and whether it is load-bearing, meaning it merges a pair no other rule had already merged. On the two cohorts with intact identifiers every non-subject rule is redundant; §S4 is unaffected, but the provenance-degradation experiment in the main text exists precisely because that redundancy disappears when identifiers decay.

S10 Audit gate: blocking, reported and specified-but-unenforced checks

Table S8 separates what the released gate enforces from what it reports and from what the framework specifies but does not yet enforce. The distinction matters because an audit whose advertised

Table S7: Which edge rules actually fire, per dataset, counted from the released graph-audit artefacts. The final column is the one that matters for interpreting the component layer: an edge is *load-bearing* only if it merges a pair that no other rule had already merged. On ADNI the session, series-UID and acquisition-date rules fire on 223, 209 and 222 pairs respectively, yet every resulting component still contains exactly one subject, so each of those edges joined a pair that `same_subject` had already joined. They are redundant on these cohorts rather than inert, and become load-bearing only where subject identity is absent, corrupted, or inferred wrongly. The main text's provenance-degradation experiment tests that claim directly by deleting subject identifiers at controlled rates.

Dataset	Edge rule	Pairs joined	Load-bearing here?
ADNI1	`same_subject`	1,800	Yes (sole component-forming rule)
ADNI1	`same_session`	223	No (subsumed by subject)
ADNI1	`same_series_uid` (source volume)	209	No (subsumed by subject)
ADNI1	`same_acq_date`	222	No (subsumed by subject)
267/382 ADNI components are formed by `same_subject` alone; in the other 115 the redundant rules also fire.			
Kaggle	`same_subject` (filename-inferred)	6,076	Fires, but the key is not participant identity
Kaggle	`exact_sha256_duplicate`	0	No pairs found
Kaggle	near-duplicate (dHash ≤ 4)	1,582	Not merged (REVIEW policy)
OASIS-1	`same_subject`	all 416 components	Yes
OASIS-1	`same_session`	all 416 components	No (subsumed by subject)
OASIS-1	`same_source_volume`	all 416 components	No (subsumed by subject)

criteria exceed its implemented criteria is itself an evaluation-integrity failure.

S11 Site and scanner association with label, per partition

Table S9 reports the per-partition association between each audited covariate and the CN/AD label. The audit surfaces the confound; SplitGuard-AD does not attempt to remove it, and the main text states why.

S12 Test-partition composition under the three protocols

Table S10 gives the number of distinct participants, images and images per participant in each protocol's test partition. The three hold image count roughly constant while differing fourfold in how many participants those images come from, which is the asymmetry every protocol-to-protocol comparison in the main text is conditioned on.

References

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Table S8: GO/NO-GO decision criteria of the released SplitGuard-AD audit (`scripts/audit_adni_splits.py`). We distinguish what the released gate *enforces* from what it *reports* and from what the framework specifies but does not yet enforce, because a leakage audit whose advertised criteria exceed its implemented criteria is itself an evaluation-integrity failure. Only the three blocking checks determine the binary decision; every other quantity is written to the audit report for author and reviewer judgement without halting the pipeline. The gate certifies partition disjointness of the split manifest, *not* site-generalisability, scanner-shift robustness, calibration, fairness, or clinical deployment readiness.

Check	**Status in v1**	**Behaviour**
Subject overlap across partitions	**Blocking**	Any non-blank subject ID in > 1 partition sets `passed=False`; blank and `unknown` IDs are skipped, so a cohort without identifiers cannot pass by vacuity.
Session overlap across partitions	**Blocking**	Any session ID in > 1 partition sets `passed=False`.
Component overlap across partitions	**Blocking**	Any component ID in > 1 partition sets `passed=False`.
Per-partition label balance	Reported	Class counts and ratios written to the report.
Scanner field-strength balance	Reported	Marginal distribution per partition.
Age and sex summaries	Reported	Per-partition demographic summary.
Site–label association (Cramér's V)	Reported	Computed by a separate script (reported with the ADNI results in the main text); not consulted by the gate.
Component-size balance across partitions	Reported	Per-partition component-size distribution, with a warning when partitions are not drawn from comparable participants. Added after auditing this project's own frozen splits, on which it fires (see the main text's Limitations); not consulted by the gate.
Unresolved subject or diagnosis identifiers	**Not enforced in v1**	No check exists. Two ADNI scans with unresolved visit-level diagnosis passed the gate for this reason (see the main text's ADNI cohort description).
Frozen-manifest hash verification	**Not enforced in v1**	Hashed manifests are produced for release but the gate does not re-verify them.
Residual near-duplicate cross-split	**Not enforced in v1**	Near-duplicate edges are applied when *building* components, not re-checked after partitioning.
Within-component label agreement	**Not enforced in v1**	Mixed-label converter components are handled by an explicit pre-specified exclusion rule (stated in the main text's ADNI cohort description) rather than by the gate.

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Table S9: Site and sex confound audit on the five ADNI primary-arm component-safe splits (1,125 manifest rows, 1,123 diagnosis-resolved; CN/AD rows only). Bias-corrected Cramér's V (Bergsma [14]) between each covariate and the binary CN/AD label per partition; mean and range across the five seeds. The site–label association is substantial in every partition and grows as the partition shrinks; it is *exposed* by the audit but not *resolved* by the component-safe split, which is the central caveat of the main text's Limitations. The sex–label association is negligible in train and test but reaches $V = 0.34$ in the small validation partition, an instability we attribute to its size ($n = 165$) rather than to a population effect. Scanner field strength is constant (1.5 T) across every partition of this cohort, so $V(\text{scanner}|\text{label}) = 0$ by construction and the covariate is not tabulated.

Partition	Rows (mean)	Unique sites (range)	V(site|label) (mean [min–max])	V(sex|label)
Train	793	38–39	0.360 [0.347–0.375]	0.000
Val	165	8–10	0.479 [0.464–0.495]	0.343
Test	165	11–12	0.538 [0.418–0.715]	0.046

Scanner field strength is constant (1.5 T) in every partition; $V(\text{scanner}|\text{label}) = 0$ throughout.

Table S10: Test-partition composition per protocol on the ADNI converter-inclusive arm (five-seed means), the arm on which the subject-level aggregation is computed. The three protocols hold image count roughly constant but differ fourfold in how many distinct participants those images come from, because tightening the split rule packs more scans of fewer people into the test partition. Every protocol-to-protocol AUROC comparison in this paper is conditioned on this asymmetry, and we report it here rather than leaving it implicit.

Protocol	**Test participants**	**Test images**	**Images per participant**
A (random)	160.8	213.0	1.32
B (subject-only)	48.0	220.0	4.58
C (component-safe)	38.0	203.2	5.35

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